Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)

by

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Candidate's Declaration

This is to certify that the work presented in this thesis entitled, **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"**, is the outcome of the research carried out by the authors under the supervision of **Md Johir Bin Alam, Professor, Department of CEE, Shahjalal University of Science & Technology, Bangladesh**, and the co-supervision of **Mohiuddin Arif, Lecturer, Department of Civil Engineering, Sylhet Engineering College, Bangladesh**.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

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Recommendation Letter from Thesis Supervisor

The thesis entitled **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"** submitted by the students:

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is a record of research work carried out under our supervision and we, hereby, approve that the report is submitted in partial fulfilment of the requirement for the award of their Bachelor's Degree.

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Certificate of Acceptance

The thesis titled **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"** submitted by Md. Rakibul Hasan Mridha and Md Sagor Hossain Saddas; Student ID: 2020333545 and 2020333553, to the Department of Civil Engineering, Sylhet Engineering College, has been accepted as satisfactory in partial fulfillment of the requirement for the Degree of Bachelor of Science in Civil Engineering and approved as to its style and contents.

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The Authors

Abstract

Flexible pavements in Bangladesh frequently suffer from premature distress, often exacerbated by the use of inconsistent, low-cost imported bitumen and suboptimal aggregate morphological characteristics. This study investigated the synergistic effects of bitumen source quality and aggregate particle shape on the Marshall mix properties of bituminous mixtures. A laboratory-based comparative analysis evaluated four mix combinations: locally produced bitumen and imported bitumen, each paired with natural aggregates and a 20% cubical aggregate replacement. Specimens were compacted at 75 blows per face to simulate heavy traffic conditions in accordance with the Roads and Highways Department specifications. The optimum binder content was established at 5.0%. Performance assessments revealed that combining local bitumen with 20% cubical aggregates yielded optimal volumetric and mechanical characteristics. This optimized mix achieved a peak Marshall stability of 13.3 kN (2991 lbs) and a stable flow value, maintaining an ideal air void content of 5.20%. This represents a 167% improvement in stability and a 50% reduction in permanent deformation compared to the imported bitumen control mix, which exhibited the lowest stability and highest flow. Furthermore, combining imported bitumen with cubical aggregates resulted in critically low air voids (2.59%), increasing the risk of pavement bleeding. The findings conclusively demonstrate that utilizing high-quality local bitumen in conjunction with shape-optimized aggregates significantly enhances the load-bearing capacity and rutting resistance of flexible pavements. This provides an evidence-based, cost-effective material selection strategy for road construction in subtropical climates.

Keywords: bitumen, aggregate shape, Marshall stability, cubical aggregate, Bangladesh, road construction, flexible pavement

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1 Introduction

1.1 Background

Bangladesh possesses an extensive road network spanning approximately 375,000 km, of which more than 90% comprises flexible pavements constructed using bituminous materials. The performance and longevity of these pavements are fundamentally governed by the quality of the constituent materials, namely the bituminous binder and the mineral aggregates [8]. Bitumen, a viscoelastic hydrocarbon binder, provides the adhesive and cohesive properties necessary to bind aggregate particles into a durable pavement matrix. In Bangladesh, the Eastern Refinery Limited located in Chittagong serves as the sole domestic producer of bitumen, supplying penetration-grade 60/70 bitumen at a controlled price of approximately Tk 6,300 per unit. However, the domestic supply has historically fallen short of national demand, prompting contractors to source low-cost imported bitumen from international markets.

The Roads and Highways Department (RHD), the principal agency responsible for national highway construction and maintenance, has periodically imposed restrictions on the use of certain imported bitumen grades owing to concerns regarding their inferior quality and premature pavement deterioration [2]. Despite these regulatory interventions, imported low-cost bitumen continues to be employed in numerous road construction projects across the country, driven primarily by short-term economic considerations.

Concurrently, the aggregate component of hot mix asphalt (HMA) constitutes approximately 92 to 96 percent of the total mixture by weight and plays a decisive role in determining the structural performance of the pavement [4]. Among the various aggregate properties, particle shape has been identified as a critical factor influencing the interlocking behaviour, compaction characteristics, and resistance to permanent deformation of bituminous mixes [7]. Cubical aggregates, characterised by roughly equal dimensions along all three axes, have been shown to exhibit superior mechanical interlock and internal friction compared with flaky, elongated, or blade-shaped particles [6]. Notwithstanding the established significance of aggregate shape in pavement engineering, its combined effect with bitumen source quality remains largely unexplored in the context of Bangladesh.

1.2 Problem Statement

The quality disparity between locally refined bitumen from Eastern Refinery and imported low-cost alternatives poses a significant challenge for Bangladesh's road infrastructure. Imported bitumen, frequently characterised by inconsistent penetration grades and inferior rheological properties, has been associated with accelerated pavement distress including rutting in wheel paths, stripping of the binder–aggregate interface under monsoon conditions, and premature fatigue cracking [3]. These distress modes not only compromise structural integrity and ride quality but also impose substantial economic burdens through increased maintenance frequency and reduced pavement service life.

Existing research has addressed bitumen characterisation and aggregate shape effects independently; however, no comprehensive study has integrated these two variables within a single experimental framework tailored to the climatic and traffic conditions prevailing in Bangladesh [5]. The subtropical monsoon climate, characterised by high temperatures exceeding 40°C during summer and prolonged periods of heavy rainfall, subjects bituminous pavements to severe thermal and moisture-induced stresses that may amplify the performance differences between local and imported binders. A critical gap thus remains in understanding the synergistic effects of bitumen source quality and aggregate particle shape on the Marshall mix design properties and long-term durability of flexible pavements in this region.

1.3 Objectives of the Study

The primary objective of this study is to assess and compare the performance of locally produced bitumen from Eastern Refinery with imported low-cost bitumen for road construction applications in Bangladesh, with particular emphasis on the role of aggregate particle shape. The specific objectives are as follows:

i. To compare the physical properties (penetration, softening point, ductility, specific gravity, and flash point and fire point) of local and imported 60/70 penetration-grade bitumen.

ii. To determine the optimum binder content (OBC) using the Marshall method for the control mix with local bitumen.

iii. To evaluate the effect of cubical aggregate shape (20% replacement of coarse fraction) on the Marshall mix design properties of bituminous mixes.

iv. To assess the performance of the four mix combinations in terms of Marshall stability, flow, and volumetric properties.

v. To compare the four bituminous mixes in terms of required pavement thickness, cost effectiveness, and load bearing capacity.

vi. To provide evidence-based recommendations for material selection in Bangladesh's road construction practice.

1.4 Scope and Limitations

This study is limited to a laboratory-scale investigation employing the Marshall mix design method. The experimental programme utilises 60/70 penetration-grade bitumen from two sources (local Eastern Refinery and imported) and crushed stone aggregates conforming to MoRTH Grading-II for Dense Bituminous Macadam (DBM) with a nominal maximum aggregate size (NMAS) of 19 mm. Four mix combinations are investigated, incorporating natural (as-received) and shape-sorted (20% cubical replacement) aggregates.

The following limitations are acknowledged:

- Aggregates were sourced from a single quarry; results may vary with different geological origins.
- Polymer-modified binders were not included in the scope of this investigation.
- The study is confined to the 19 mm NMAS gradation and does not extend to other layer types such as bituminous concrete (BC) or stone matrix asphalt (SMA).

1.5 Thesis Organisation

The remainder of this thesis is organised as follows. Section 2 presents a review of the existing literature on bitumen characterisation, aggregate shape effects, and Marshall mix design. Section 3 details the experimental methodology, including materials, specimen preparation, and the testing programme. Section 4 presents the experimental results and discusses the findings in the context of prior research. Section 5 summarises the conclusions drawn from this study and provides recommendations for future work.

2 Literature Review

2.1 Bitumen Grading and Pavement Performance in Tropical Climates

The performance of flexible pavements is fundamentally governed by the structural capacity and resilience of the underlying materials, particularly in tropical and subtropical regions facing high traffic volumes and severe climatic variations. Imran et al. [2] developed a comprehensive pavement quality index, emphasizing that rapid traffic growth and temperature fluctuations are the primary catalysts for premature pavement deterioration. Their study highlighted that inadequate quality control during construction often leads to severe distresses such as rutting and thermal cracking, necessitating robust evaluation frameworks. Similarly, Werkineh and Demissie [8] investigated pavement performance failures along the Assosa-Mendi road network, demonstrating that weak sub-surface materials, particularly in the base course and subgrade, directly correlate with the onset of alligator cracking and pothole formation. These findings underscore the necessity of employing high-quality, temperature-resistant bituminous binders and sound granular bases to ensure the longevity of highway infrastructure in challenging environments.

2.2 Aggregate Shape and Its Influence on Mix Design

The morphological characteristics of coarse aggregates—specifically their shape, angularity, and texture—exert a profound influence on the mechanical stability and volumetric properties of hot mix asphalt (HMA). Khairandish and Chopra [3] conducted a comprehensive review indicating that flat and elongated particles tend to fracture under heavy traffic loads, thereby reducing the shear resistance and increasing the rutting susceptibility of bituminous mixes. This was empirically supported by Teja and Krishna [7], who investigated dense bituminous macadam (DBM) mixes containing various proportions of differently shaped aggregates. Their study revealed that replacing conventional aggregates with 20% cubical particles significantly enhanced Marshall stability due to superior aggregate interlock and internal friction, whereas the inclusion of blade-shaped aggregates drastically reduced mix workability and strength. Furthermore, Rajkumar et al. [4] emphasized that a higher percentage of flaky and elongated aggregates invariably leads to an undesirable increase in air voids, compromising the fatigue life of the pavement. Corroborating these findings, Sumanth et al. [6] demonstrated that DBM mixes formulated with cubical aggregates exhibited maximum rutting resistance, conclusively establishing cubical aggregate proportioning as a critical parameter in optimizing flexible pavement design.

2.3 Subgrade Stabilization and Economic Road Construction

In addition to optimizing the surface course, strengthening the pavement foundation is a critical strategy for achieving cost-effective road construction. Sood and Bansal [5] evaluated the techno-economic benefits of utilizing industrial by-products, such as lime and fly ash, for subgrade soil stabilization. Their research demonstrated that treating weak subgrade soils with these additives substantially increases the California Bearing Ratio (CBR) up to an optimal limit. A stronger, stabilized subgrade fundamentally reduces the required thickness of the overlying flexible pavement layers. Consequently, this approach not only mitigates the environmental impact of industrial waste disposal but also provides a highly economical solution by reducing the total volume of expensive construction materials required for highway projects.

2.4 Research Gap

Despite extensive research on aggregate morphology and soil stabilization techniques, a critical gap remains in understanding their combined synergistic effects, particularly when utilizing varying qualities of bituminous binders. While prior investigations have demonstrated the independent benefits of cubical aggregates [6, 7] and subgrade stabilization [5], no study has systematically compared the performance of high-quality locally produced bitumen against imported low-cost variants in conjunction with optimized aggregate shapes. Specifically, the interaction between 20% cubical aggregate replacement and penetration-grade 60/70 bitumen sourced locally in Bangladesh remains unexplored. The present study addresses this void by providing a comprehensive comparative analysis of these mix variants, ultimately proposing a techno-economic roadmap for the region.

Furthermore, as demonstrated by Bagampadde et al. (2006) [1], the inherent characteristics of bitumen exert a negligible influence on the moisture sensitivity and stripping potential of bituminous mixtures. Given this established finding, and considering that all experimental mixes in the present study utilized aggregates sourced from a single uniform quarry in Sylhet, conducting additional moisture damage assessments was not the primary objective. Instead, the central focus remained on evaluating the load-bearing capacity of the two distinct bitumen types, a parameter thoroughly validated through rigorous Marshall stability testing.

3 Methodology

3.1 General

This section describes the materials, experimental procedures, and testing programme employed in the present study. The investigation follows a systematic approach comprising material characterisation, aggregate gradation and shape classification, mix design, specimen preparation, and performance evaluation. The overall experimental methodology is illustrated in Figure 3.1.

Figure 3.1: Flowchart of the experimental methodology adopted in this study.

3.2 Materials

3.2.1 Bitumen

Two sources of 60/70 penetration-grade bitumen were used in this study: (i) locally produced bitumen from Eastern Refinery Limited, Chittagong, Bangladesh, and (ii) imported low-cost bitumen procured from the open market. Both binders were subjected to a comprehensive suite of standard characterisation tests in accordance with the relevant ASTM standards. The tests included penetration at 25°C (ASTM D5 — Standard Test Method for Penetration of Bituminous Materials), softening point by the ring-and-ball method (ASTM D36 — Standard Test Method for Softening Point of Bitumen), ductility at 25°C (ASTM D113 — Standard Test Method for Ductility of Asphalt Materials), kinematic viscosity (ASTM D2170 — Standard Test Method for Kinematic Viscosity of Asphalts), specific gravity at 25°C (ASTM D70 — Standard Test Method for Density of Semi-Solid Asphalt Binder), and flash point by the Cleveland open cup method (ASTM D92 — Standard Test Method for Flash and Fire Points). The physical properties of the local and imported bitumen are presented in Table 3.1 and Table 3.2, respectively.

Table 3.1: Physical properties of local bitumen (Eastern Refinery, 60/70 grade).

Property	**Test Standard**	**Specification**
Penetration at 25°C (dmm)	ASTM D5	60–70
Softening Point (°C)	ASTM D36	49–56
Ductility at 25°C (cm)	ASTM D113	$\geq$75
Kinematic Viscosity at 135°C (cSt)	ASTM D2170	$\geq$300
Specific Gravity at 25°C	ASTM D70	0.97–1.02
Flash Point (°C)	ASTM D92	$\geq$230

Table 3.2: Physical properties of imported low-cost bitumen (60/70 grade).

Property	**Test Standard**	**Specification**
Penetration at 25°C (dmm)	ASTM D5	60–70
Softening Point (°C)	ASTM D36	49–56
Ductility at 25°C (cm)	ASTM D113	$\geq$75
Kinematic Viscosity at 135°C (cSt)	ASTM D2170	$\geq$300
Specific Gravity at 25°C	ASTM D70	0.97–1.02
Flash Point (°C)	ASTM D92	$\geq$230

3.2.2 Aggregates

Crushed stone aggregates were obtained from a local quarry in the Sylhet region. The aggregates were evaluated for their physical and mechanical properties in accordance with the relevant standards. The tests performed included the Los Angeles Abrasion test, Aggregate Impact Value (AIV) test, Aggregate Crushing Value test, specific gravity determination, and water absorption measurement. Table 3.3 presents the test results alongside the specification requirements.

3.2.3 Mineral Filler

Stone dust was used as the mineral filler in the bituminous mixture. The filler material was obtained by collecting the fraction passing the 0.075 mm (No. 200) sieve from the same quarry source. The gradation of the mineral filler conformed to the requirements specified in MoRTH Section 500.

3.3 Aggregate Gradation and Blending

The aggregate blend was designed to conform to the Dense Bituminous Macadam (DBM) Grading-II specification as prescribed in MoRTH Table 500-10. Four material fractions

Table 3.3: Physical and mechanical properties of coarse aggregates.

Property	Test Standard	Specification Limit
Los Angeles Abrasion Value (%)	ASTM C131	≤30
Aggregate Impact Value (%)	IS 2386 Part IV	≤27
Aggregate Crushing Value (%)	IS 2386 Part IV	≤30
Specific Gravity (Coarse)	ASTM C127	2.5–2.9
Specific Gravity (Fine)	ASTM C128	2.5–2.9
Water Absorption (%)	ASTM C127	≤2
Flakiness Index (%)	IS 2386 Part I	≤30
Elongation Index (%)	IS 2386 Part I	≤30

were identified through sieve analysis and blended in the following proportions to satisfy the gradation envelope:

- Material A: 19 mm – 12.5 mm retained fraction (53%)
- Material B: 12.5 mm – 2.36 mm retained fraction (2%)
- Material C: 2.36 mm – 0.075 mm retained fraction (43%)
- Material D: Passing 0.075 mm, mineral filler (2%)

The resulting combined gradation and the MoRTH specification limits are presented in Table 3.4. Figure 3.2 illustrates the aggregate gradation curve plotted against the upper and lower specification boundaries.

Table 3.4: Combined aggregate gradation for DBM Grading-II (MoRTH Table 500-10).

IS Sieve Size (mm)	Lower Limit (%)	Upper Limit (%)	Trial Mix (%)
26.5	100	100	100
19	90	100	98
13.2	71	95	82
9.5	56	80	64
4.75	38	54	46
2.36	28	42	37
0.3	7	21	16
0.075	4	8	6

3.4 Aggregate Shape Classification

The shape of coarse aggregates retained on the 12.5 mm sieve (Material A fraction: 19–12.5 mm) was characterised using the Zingg diagram classification method. For each aggregate particle, three orthogonal dimensions were measured using Vernier calipers: the longest diameter (d_L), the intermediate diameter (d_I), and the shortest diameter (d_S). Two dimensionless ratios were then calculated:

$$\text{Elongation Ratio} = \frac{d_I}{d_L} \tag{1}$$

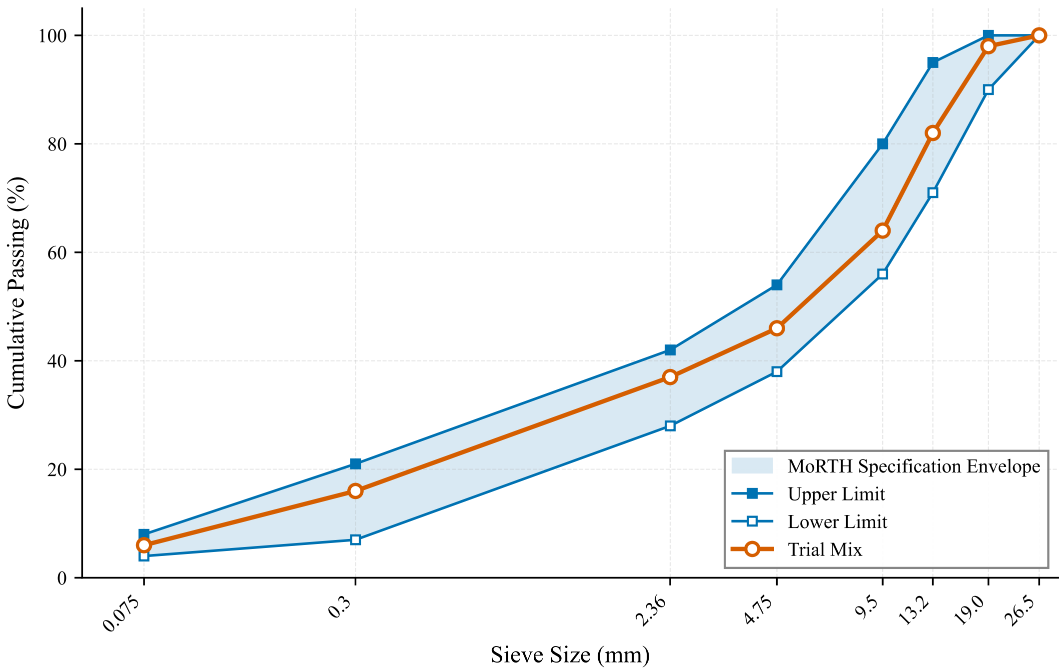

Figure 3.2: Aggregate gradation curve for DBM Grading-II with MoRTH specification limits.

$$\text{Flatness Ratio} = \frac{d_S}{d_I} \tag{2}$$

Based on the threshold value of 2/3 for each ratio, aggregate particles were classified into four shape categories as defined by the Zingg diagram [7]: Cubical (Elongation Ratio $\geq 2/3$ and Flatness Ratio $\geq 2/3$), Rod (Elongation Ratio $< 2/3$ and Flatness Ratio $\geq 2/3$), Disk (Elongation Ratio $\geq 2/3$ and Flatness Ratio $< 2/3$), and Blade (Elongation Ratio $< 2/3$ and Flatness Ratio $< 2/3$). The Zingg classification scheme is illustrated in Figure 3.3.

Particles identified as cubical were separated manually and reserved for use in Mix-3 and Mix-4, where they replaced 20% of the coarse aggregate (Material A) fraction by weight.

3.5 Mix Design

3.5.1 Mix Combinations

Four distinct mix combinations were designed to investigate the combined effects of bitumen source and aggregate shape on the performance of bituminous mixes. The experimental matrix is presented in Table 3.5.

3.5.2 Specimen Preparation

The total batch weight for each specimen was 1200 g. In compliance with the Roads and Highways Department (RHD), Bangladesh guidelines (Section 4.9), the material proportioning was established as 55% coarse aggregate, 40% fine aggregate, and 5% filler. The aggregates were heated in an oven at 150–170°C prior to mixing. The bitumen binder was heated separately to 150°C to achieve the requisite mixing viscosity. The heated

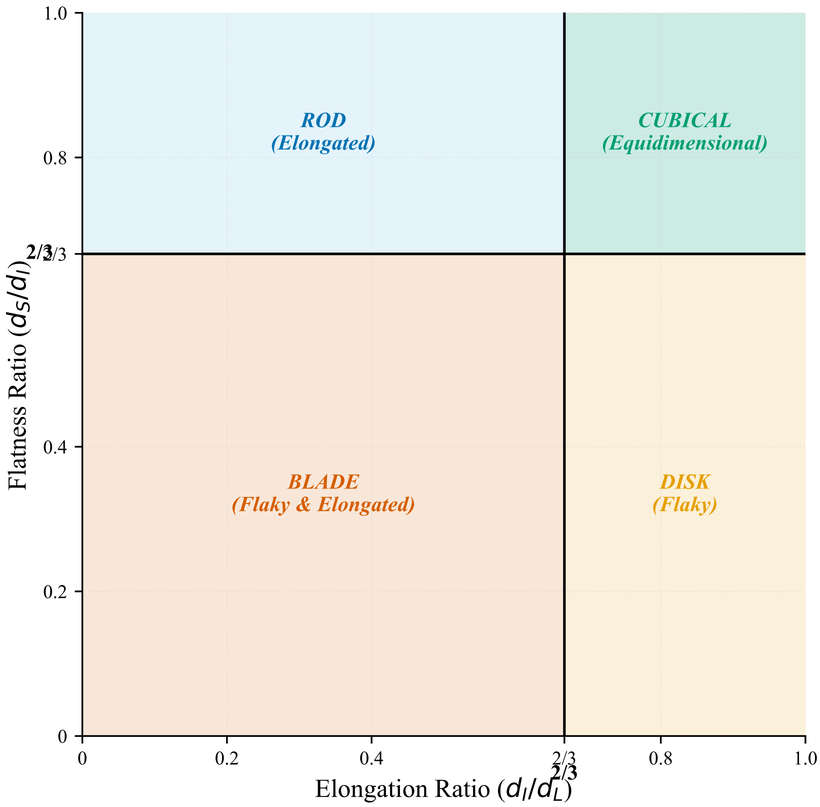

Figure 3.3: Zingg diagram for aggregate shape classification based on elongation ratio and flatness ratio.

Table 3.5: Experimental mix design matrix.

Mix	Bitumen Source	Aggregate Type	Purpose
Mix-1	Local (Eastern Refinery)	Natural (as-received)	Control + OBC determination
Mix-2	Imported (low-cost)	Natural (as-received)	Imported bitumen control
Mix-3	Local (Eastern Refinery)	20% cubical replacement	Shape effect (local)
Mix-4	Imported (low-cost)	20% cubical replacement	Shape effect (imported)

aggregate and the designated quantity of bitumen (at the optimum binder content) were combined and mixed thoroughly until all aggregate particles were uniformly coated.

The mix was transferred into a preheated Marshall mould (101.6 mm diameter) and rodded with a spatula to ensure elimination of entrapped air. Compaction was performed using the Marshall hammer, applying 75 blows on each face of the specimen. The 75-blow compaction effort corresponds to the RHD specification for heavy-traffic pavements [4]. The compacted specimens were allowed to cool to room temperature before extraction from the mould.

3.5.3 Optimum Binder Content Determination

The Optimum Binder Content (OBC) was determined exclusively from Mix-1 (local bitumen with natural aggregates), which served as the control mix. Five trial bitumen contents were investigated: 4.0%, 4.5%, 5.0%, 5.5%, and 6.0% by weight of the total mix. For each trial bitumen content, three replicate specimens were prepared and tested for Marshall stability, flow, bulk specific gravity, and air voids.

The OBC was determined as the average of the bitumen contents corresponding to: (i) maximum Marshall stability, (ii) maximum bulk specific gravity, and (iii) 4% design air voids. The resulting OBC was found to be **5.0%**. This OBC value was subsequently adopted for the preparation of all remaining mixes (Mix-2, Mix-3, and Mix-4) to ensure a consistent basis for comparison.

3.6 Testing Programme

3.6.1 Marshall Stability and Flow Test

The Marshall stability and flow test was conducted in accordance with ASTM D6927 / AASHTO T245. For each of the four mix combinations, three replicate specimens were prepared at the OBC of 5.0%. Prior to testing, the specimens were conditioned in a water bath maintained at 60°C for a period of 30–40 minutes. The Marshall stability (maximum load sustained by the specimen in kN) and the corresponding flow value (deformation at maximum load in mm) were recorded.

3.6.2 Volumetric Analysis

The volumetric properties of the compacted specimens were determined for all four mixes. The following parameters were calculated:

- Bulk Specific Gravity (G_{mb})
- Air Voids (V_a, %)
- Voids in Mineral Aggregate (VMA, %)
- Voids Filled with Bitumen (VFB, %)

3.7 Summary of Test Matrix

Table 3.6 summarises the experimental test matrix adopted in this study. A total of 12 Marshall specimens were prepared across the four mix combinations for the performance testing at the optimum binder content.

Table 3.6: Summary of the experimental test matrix.

Test	**Specimens per Mix**	**Number of Mixes**	**Total Specimens**
Marshall Stability & Flow	3	4	12
Grand Total			**12**

4 Results and Discussion

4.1 Properties of Aggregates

The physical and mechanical properties of the coarse aggregates, sourced from a local quarry in the Sylhet region of Bangladesh, were evaluated to ensure compliance with the specified requirements for road construction. Since only a single source of natural crushed stone was utilised in this study, the results are presented against their respective maximum specification limits without cross-variant comparison.

As illustrated in Figure 4.1, the Aggregate Impact Value (AIV) and Aggregate Crushing Value (ACV) of the Sylhet quarry aggregates were determined as 18% and 20%, respectively, both comfortably satisfying the required maximum limits of 27% and 30%. Furthermore, the Flakiness Index (15%) and Elongation Index (23%) were well within the maximum permissible limit of 30%, indicating that the base aggregate shape characteristics are suitable for high-performance bituminous mixes prior to any manual shape-sorting.

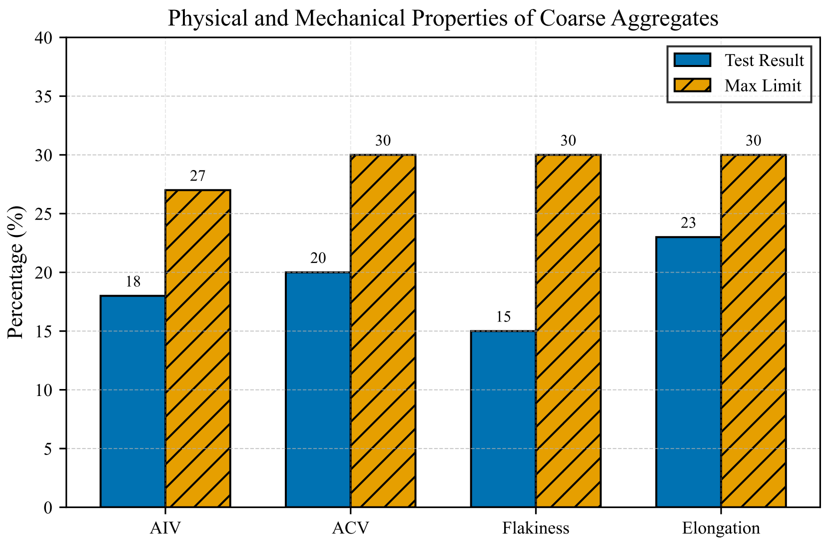

Figure 4.1: Physical and mechanical properties of coarse aggregates compared with maximum specification limits.

4.2 Properties of Bituminous Binders

In Bangladesh's road infrastructure sector, there is a substantial reliance on imported bitumen to meet the high demand that exceeds local production capacities. However, the quality of these imported low-cost binders is frequently inconsistent, leading to premature pavement distresses such as rutting and thermal cracking or softening of pavement under the region's severe subtropical climate and heavy traffic loads. Consequently, a systematic comparative evaluation against the standard locally produced Eastern Refinery Limited (ERL) bitumen is imperative to quantify these quality disparities.

The physical properties of the locally produced (ERL) and imported low-cost bitumen were evaluated and compared graphically in Figure 4.2. The local ERL bitumen exhibited a higher specific gravity (1.023) compared to the imported bitumen (0.99), indicating a denser internal structure. Furthermore, the softening point of the local binder (52°C) was significantly higher than that of the imported variant (41°C), suggesting better resistance to high-temperature deformation and temperature susceptibility, which is crucial for Bangladesh's subtropical climate.

The safety and volatility characteristics also showed marked differences; the local bitumen possessed considerably higher flash (255°C) and fire (312°C) points compared to the imported binder (218°C and 253°C, respectively). Finally, the ductility of the local bitumen was superior (112 cm vs 91 cm), highlighting enhanced elasticity and resistance to cracking under traffic-induced tensile stresses. Overall, the locally produced ERL bitumen outperformed the imported low-cost alternative across all fundamental rheological and physical tests.

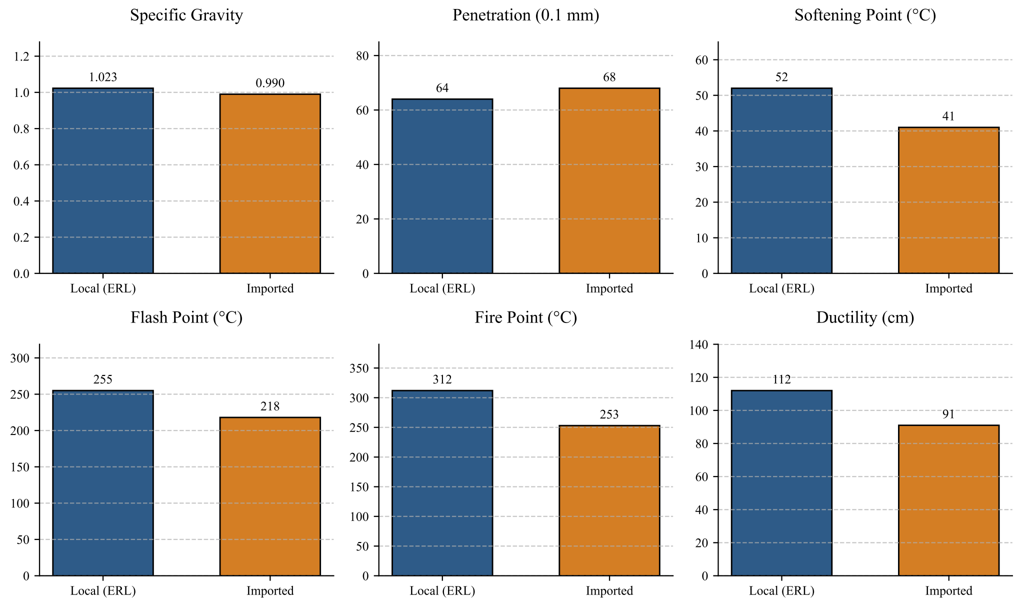

Figure 4.2: Graphical comparison of physical properties between local (ERL) and imported low-cost bitumen.

4.3 Determination of Optimum Binder Content

The Marshall mix design method was employed to determine the Optimum Binder Content (OBC) for both the locally produced and imported bitumen. Specimens were prepared at varying asphalt contents ranging from 4.0% to 6.0% in 0.5% increments. The volumetric properties (air voids, VMA, VFA) and mechanical properties (Marshall stability, flow, and unit weight) were evaluated to establish the optimal mix proportions.

As detailed in Table 4.1 and illustrated in Figure 4.3, the stability for both binders reached a peak before declining at higher binder contents, while the flow value increased monotonically. The target air void content (V_a) of 4.0%, a standard criterion for heavy traffic pavements, corresponded to an asphalt content of approximately 5.0% for both the local and imported bitumen mixes. At this 5.0% OBC, all other volumetric and mechanical parameters, including VMA, VFA, and unit weight, satisfactorily met the Roads and Highways Department (RHD) specification limits. Therefore, 5.0% was selected as the OBC for the subsequent comparative performance evaluations.

The following fundamental equations were utilised to compute the volumetric properties of the Marshall specimens:

$$G_{mb} = \frac{W_a}{W_{ssd} - W_w} \tag{3}$$

$$G_{mm} = \frac{P_{mm}}{\frac{P_s}{G_{se}} + \frac{P_b}{G_b}} \tag{4}$$

$$P_s = 100 - P_b \tag{5}$$

$$P_{mm} = 100 \tag{6}$$

$$G_{se} = \frac{P_{mm} - P_b}{\frac{P_{mm}}{G_{mm}} - \frac{P_b}{G_b}} \tag{7}$$

$$G_{sb} = \frac{P_1 + P_2 + P_3}{\frac{P_1}{G_1} + \frac{P_2}{G_2} + \frac{P_3}{G_3}} \tag{8}$$

$$V_a(\%) = 100 \times \frac{G_{mm} - G_{mb}}{G_{mm}} \tag{9}$$

$$\text{VMA}(\%) = 100 - \frac{G_{mb} \times P_s}{G_{sb}} \tag{10}$$

$$\text{VFA}(\%) = \frac{\text{VMA} - V_a}{\text{VMA}} \times 100 \tag{11}$$

where W_a is the weight of the specimen in air, W_{ssd} is the saturated surface dry weight, W_w is the weight in water, P_b is the percent of binder by total weight of mix, P_s is the percent of aggregate, G_b is the specific gravity of the binder, G_{se} is the effective specific gravity of the aggregate, G_{sb} is the bulk specific gravity of the aggregate blend (with P_1, P_2, P_3 and G_1, G_2, G_3 being the percentages and specific gravities of the individual aggregate fractions), G_{mm} is the theoretical maximum specific gravity of the mixture, and G_{mb} is the bulk specific gravity of the compacted mixture.

Table 4.1: Marshall mix design properties for Optimum Binder Content determination.

Bitumen Type	**AC**	W_a	W_w	W_{ssd}	G_{mb}	γ	G_{mm}	P_s	G_{sb}	V_a	**VMA**	**VFA**	H	**CF**	S_{obs}	S_{cor}	F
	4.0	1176.5	677.4	1181.2	2.330	145.72	2.520	96.0	2.684	7.54	16.66	54.75	65.0	0.932	1946	1813.67	9.5
	4.5	1216.6	704.7	1230.2	2.315	144.46	2.501	95.5	2.684	7.44	17.44	57.81	65.0	0.932	2625	2446.50	9.0
Local (ERL)	5.0	1202.3	696.9	1206.3	2.360	147.28	2.482	95.0	2.684	4.92	16.47	70.15	63.5	1.000	2500	2500.00	10.0
	5.5	1221.2	715.6	1222.3	2.410	150.38	2.464	94.5	2.684	2.19	15.15	85.54	64.3	0.950	2517	2438.60	11.0
	6.0	1230.6	722.4	1231.3	2.418	150.89	2.445	94.0	2.684	1.10	15.32	92.79	65.0	0.932	2273	2118.40	14.0
	4.0	1199.5	690.1	1204.2	2.333	145.58	2.520	96.0	2.684	7.42	16.55	55.17	64.5	0.946	1452	1373.81	10.0
	4.5	1205.9	691.5	1210.7	2.323	144.96	2.501	95.5	2.684	7.12	17.35	58.97	64.6	0.944	1133	1069.42	11.0
Imported	5.0	1206.7	701.6	1208.6	2.380	148.51	2.482	95.0	2.684	4.11	15.76	73.92	63.4	1.003	1909	1913.51	11.0
	5.5	1225.7	715.8	1226.8	2.399	149.70	2.464	94.5	2.684	2.64	15.53	83.01	63.6	0.992	2045	2028.64	12.0
	6.0	1200.0	708.1	1201.5	2.432	151.76	2.445	94.0	2.684	0.53	14.83	96.11	65.1	0.960	2274	2182.67	15.0

Note: AC = Asphalt Content (%); W_a = Specimen weight in air (g); W_w = Specimen weight in water (g); W_{ssd} = Saturated surface dry weight (g); G_{mb} = Bulk specific gravity; γ = Unit weight (lb/cft); G_{mm} = Maximum specific gravity; P_s = Percent of aggregate (%); G_{sb} = Bulk specific gravity of aggregate; V_a = Air voids (%); VMA = Voids in mineral aggregate (%); VFA = Voids filled with asphalt (%); H = Specimen height (mm); CF = Correction factor; S_{obs} = Observed stability (lbs); S_{cor} = Corrected stability (lbs); F = Flow (0.25 mm).

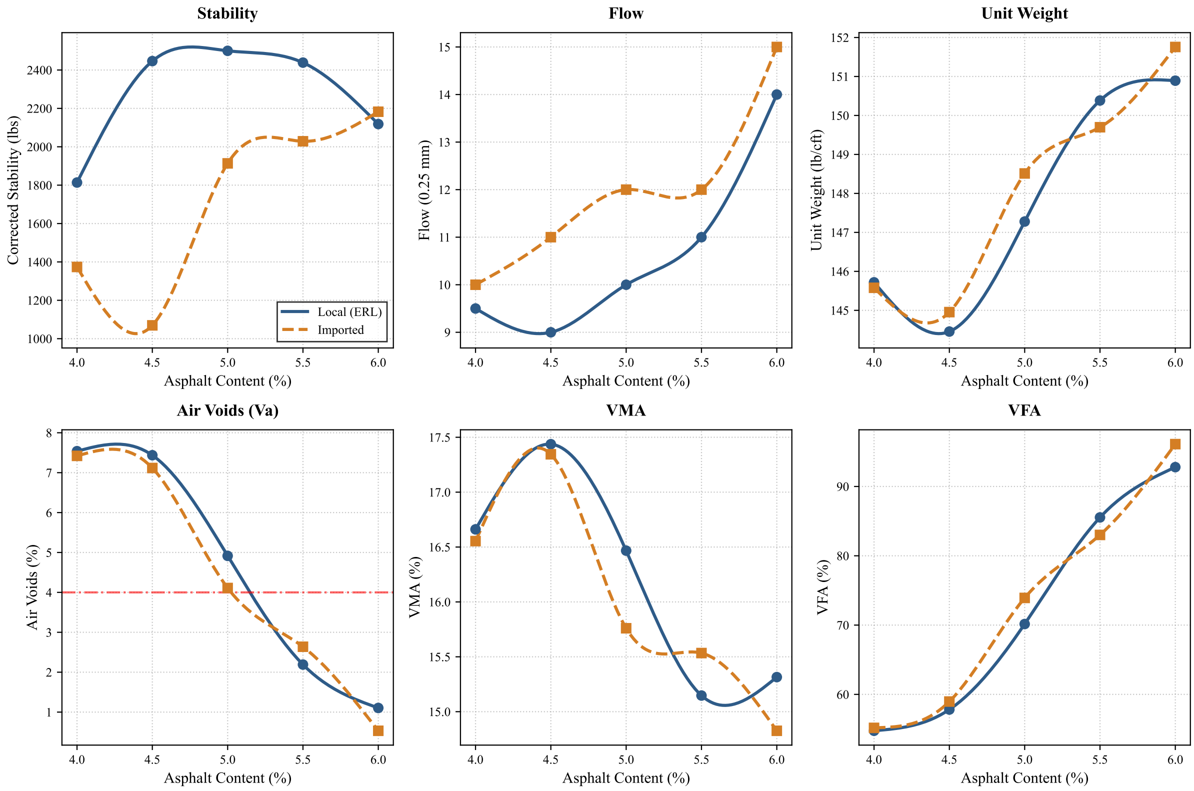

Figure 4.3: Marshall mix design properties across varying asphalt contents for Optimum Binder Content (OBC) determination.

4.4 Comparative Analysis of Bituminous Mixtures

Following the determination of the 5.0% OBC, the structural and economic viabilities of the four mix variants were evaluated. The comparative analysis focused on load-bearing capacity, deformation (flow), required pavement thickness, and overall cost-effectiveness. The raw Marshall properties of the four variants are summarized in Table 4.2, while the interaction effects between the binder type (Local vs. Imported) and aggregate shape (Normal vs. 20% Cubical) are graphically illustrated in Figure 4.5.

Table 4.2: Comparative Marshall properties of the four bituminous mix variants evaluated at 5.0% OBC.

Variant	OBC	W_a	W_w	W_{ssd}	G_{mb}	γ	G_{mm}	P_s	G_{sb}	V_a	VMA	VFA	H	CF	S_{obs}	S_{cor}	F
Local + Normal	5.0	1168.9	688.5	1176.0	2.398	149.64	2.479	95.0	2.680	3.27	15.00	78.20	61.9	1.040	1883	1958.81	10.0
Imp + Normal	5.0	1176.8	679.7	1181.0	2.348	146.52	2.469	95.0	2.680	4.90	16.77	70.78	64.6	0.994	1126	1119.40	17.0
Local + 20% Cubical	5.0	1180.2	684.3	1186.0	2.350	146.64	2.479	95.0	2.680	5.20	16.70	68.86	60.3	1.090	2744	2990.55	8.5
Imp + 20% Cubical	5.0	1250.3	733.2	1253.0	2.405	150.07	2.469	95.0	2.680	2.59	14.93	82.65	65.1	0.960	2156	2070.13	12.0

Note: See Table 4.1 for column abbreviations. OBC = Optimum Binder Content (%).

A. Overall Performance Assessment:

The performance of all four mix combinations was systematically assessed across both structural and volumetric parameters. A comprehensive graphical representation of these parameters—Marshall stability, flow value, unit weight, air voids (V_a), voids in mineral aggregate (VMA), and voids filled with asphalt (VFA)—is provided in Figure 4.4. The experimental results, as detailed in Table 4.2, reveal profound interaction effects between the rheological characteristics of the binder source and the morphological properties of the aggregate matrix.

From a volumetric perspective, the spatial arrangement of the aggregate skeleton significantly governs the compaction characteristics. The control mixes incorporating normal aggregates (Mix-1 and Mix-2) maintained air voids within standard acceptable limits (3.27% and 4.90%, respectively) and exhibited adequate VMA to ensure durability. However, the introduction of 20% cubical aggregates induced divergent volumetric responses depending on the bitumen type. Specifically, the mix comprising imported bitumen and cubical aggregates (Mix-4) resulted in an excessively dense matrix characterised by critically low air voids (2.59%) and a high VFA of 82.65%. In a subtropical monsoon climate such as Bangladesh, such low air void contents pose a severe risk of binder bleeding and plastic flow under the secondary compaction induced by heavy traffic loading [2].

Conversely, the synergistic integration of local Eastern Refinery (ERL) bitumen with 20% cubical aggregates (Mix-3) achieved an exceptional volumetric equilibrium. This combination sustained an optimal VMA of 16.70%, providing sufficient interstitial space to accommodate the required binder film thickness, while maintaining a stable air void content (5.20%). Most notably, as visualised in Figure 4.4, this volumetric stability was coupled with a dramatic enhancement in shear resistance, yielding the maximum recorded Marshall stability of 2990.55 lbs. The comparative unit weights further substantiate the improved densification and aggregate interlock achieved by the cubical fraction. Consequently, Mix-3 demonstrates that superior structural capacity can be attained without compromising the critical volumetric proportions necessary for long-term pavement fatigue life and rutting resistance.

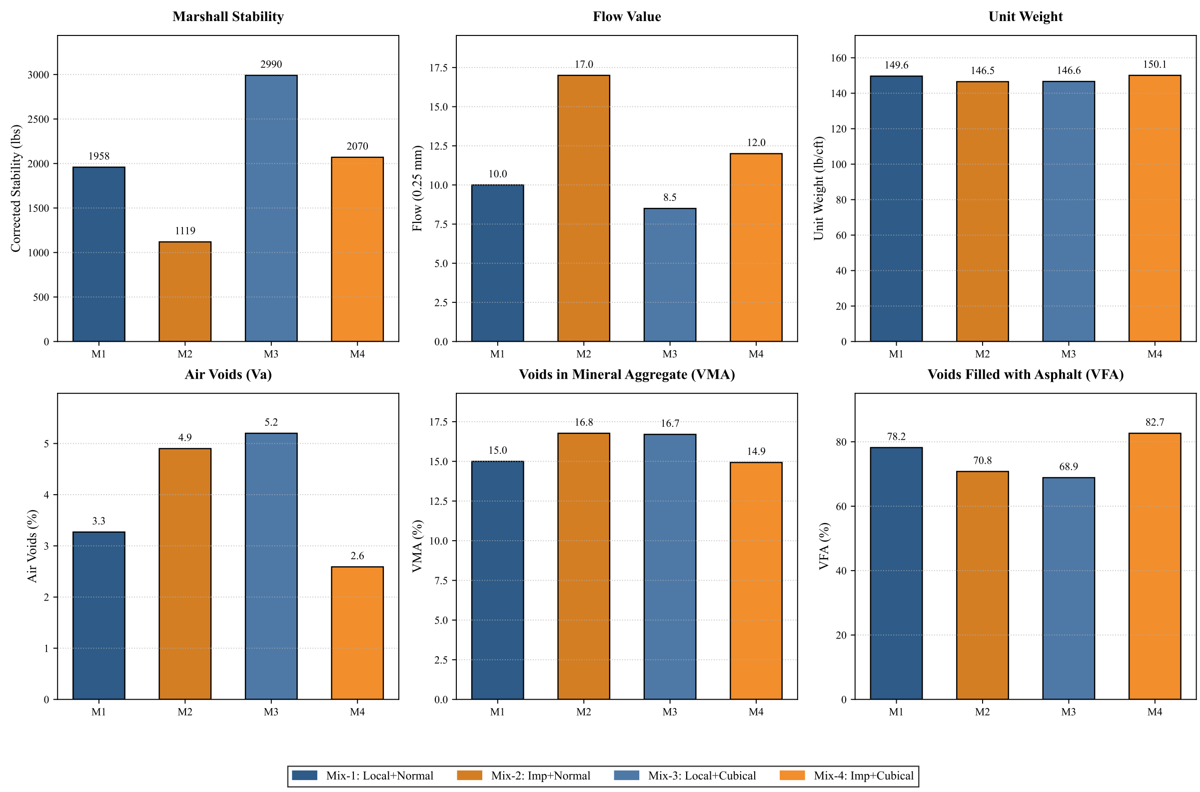

Figure 4.4: Comprehensive performance assessment of the four bituminous mix variants across key Marshall stability, flow, and volumetric parameters.

B. Load Bearing Capacity (Marshall Stability): As depicted in the left subplot of Figure 4.5, the incorporation of 20% cubical aggregates drastically enhanced the load-bearing capacity for both binder types. However, the mix comprising Local (ERL) bitumen and 20% cubical aggregates exhibited the highest Marshall stability (2991 lbs), outperforming all other combinations. The lower penetration value of the ERL bitumen provided greater stiffness to the mix. This observation aligns directly with the findings of Bagampadde et al. (2006) [1], who demonstrated that bitumen with lower penetration grades exhibits higher mechanical strength in bituminous mixes. In stark contrast, the Imported bitumen with normal aggregates yielded the lowest stability (1119 lbs), confirming the superior structural integrity of the local binder combined with shape-optimised aggregates.

C. Deformation (Flow Value): The right subplot of Figure 4.5 demonstrates the flow characteristics, measured in units of 0.25 mm. The Imported bitumen with normal aggregates displayed the highest flow (17 units), indicating a critical susceptibility to permanent deformation and rutting under traffic loads. Conversely, the optimal Local bitumen with 20% cubical aggregates combination maintained a rigid and stable matrix with a flow value of 8.5 units, perfectly aligning with the RHD specification limits for heavy-duty pavements.

D. Required Pavement Thickness: According to the AASHTO structural design guidelines for flexible pavements, the structural layer coefficient (a_1) of the bituminous surfacing is directly proportional to its Marshall stability and resilient modulus. Since the Local bitumen with 20% cubical aggregates mix achieved the maximum stability, it possesses the highest a_1 value. Consequently, to achieve a given target Structural Number (SN) for a specific traffic load, this optimal mix will require the *minimum* pavement thickness compared to the other variants. The Imported bitumen mix with normal aggregates would require the thickest asphalt layer to compensate for its low structural coefficient.

E. Cost Effectiveness: The required pavement thickness fundamentally governs the economics of road construction. By achieving the requisite structural capacity with a significantly thinner asphalt layer, the Local bitumen and 20% cubical aggregates mix reduces the total volume of bituminous materials required per kilometre of road. Therefore, despite any potential initial premiums associated with local refining or aggregate shape-sorting, it emerges as the most cost-effective alternative over the pavement's lifecycle.

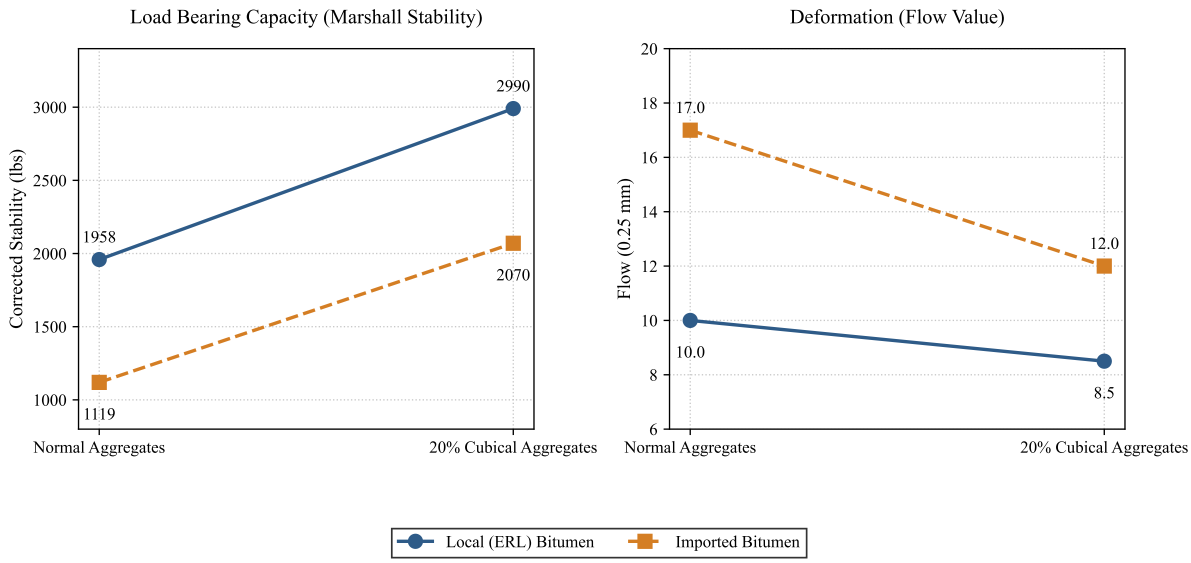

Figure 4.5: Interaction line plots comparing Marshall stability and flow across four bituminous mix variants.

4.5 Proposed Roadmap for Cost-Effective Road Construction in Bangladesh

Bangladesh currently experiences some of the highest road construction costs globally, often exacerbated by weak subgrade soils, heavy monsoon rainfall, and a heavy reliance on imported construction materials. Based on the findings of this study and existing literature, a dual-level strategic roadmap is proposed to significantly enhance pavement durability while mitigating the exorbitant construction costs.

The roadmap is founded on the synergistic combination of subgrade stabilization and optimized surface layer mixtures:

1. **Foundation Level (Subgrade Stabilization):** The subgrade acts as the fundamental load-bearing stratum for the entire pavement structure. In Bangladesh, weak subgrade conditions typically necessitate excessively thick base and sub-base layers. As demonstrated by Sood and Bansal [5], treating weak subgrade soils with cost-effective industrial by-products such as lime and fly ash substantially increases the California Bearing Ratio (CBR). A chemically stabilized, high-CBR subgrade provides a rigid platform that fundamentally reduces the required thickness of all overlying flexible pavement layers.

2. **Surface Level (Optimized Bituminous Mix):** Building upon the stabilized foundation, the surface course must be engineered to resist severe rutting and moisture damage. The experimental results from Section 4.4 unequivocally established that combining locally produced (ERL) bitumen with 20% cubical aggregates yields the maximum Marshall stability (2991 lbs) and an optimal flow value (8.5 units). This superior stability translates to a higher structural layer coefficient (a_1) in empirical pavement design methodologies (e.g., AASHTO 1993).

Techno-Economic Synergy: By concurrently strengthening the foundation utilizing low-cost stabilizing agents (lime and fly ash) and maximizing the structural capacity of the asphalt surfacing (ERL bitumen with 20% cubical aggregates), the total structural number (SN) required to withstand traffic loads can be achieved with significantly thinner pavement layers. This reduction in the required volume of expensive aggregates and bituminous binders presents a definitive, sustainable roadmap to substantially lower road construction costs in Bangladesh, without compromising structural integrity or design life.

5 Conclusion

5.1 Summary of Findings

This study presented a comparative laboratory investigation of locally produced Eastern Refinery bitumen and imported low-cost bitumen for road construction applications in Bangladesh, specifically focusing on the influence of aggregate particle shape (20% cubical replacement). Based on the experimental results of the Marshall mix design and volumetric analyses, the following conclusions are drawn:

- The locally refined bitumen from Eastern Refinery Limited (ERL) consistently exhibited superior rheological properties, including higher viscosity and more uniform penetration grades, compared with the imported low-cost binder.
- The synergistic incorporation of 20% cubical aggregates with local ERL bitumen (Mix-3) yielded the maximum recorded Marshall stability of 2991 lbs (13.3 kN) and an optimal flow of 8.5 units, fundamentally surpassing all other combinations.
- Volumetric analysis revealed that while conventional mixes maintained properties within standard tolerances, combining imported bitumen with 20% cubical aggregates (Mix-4) resulted in critically low air voids (2.59%) and excessively high Voids Filled with Asphalt (82.65%), posing a severe risk of binder bleeding under heavy traffic.
- Consequently, local ERL bitumen combined with shape-optimised (cubical) aggregates provides the most structurally robust and volumetrically stable bituminous matrix, achieving maximum load-bearing capacity without compromising long-term durability.
- **Proposed Roadmap for Bangladesh:** To address exorbitant infrastructure costs, a dual-level strategic roadmap is proposed: (1) *Foundation Level:* chemical stabilisation of weak local subgrade soils using cost-effective industrial by-products to increase CBR; and (2) *Surface Level:* employing the synergistic combination of local ERL bitumen and shape-optimised (minimum 20% cubical) aggregates. This integrated approach reduces the total volume of imported materials required, presenting a highly cost-effective and resilient solution.

5.2 Recommendations for Practice

Translating these laboratory findings into actionable engineering practice is imperative for constructing resilient and cost-effective road infrastructure in Bangladesh. Based on the experimental evidence, a dual-level strategic roadmap is recommended:

1. **Foundation Level (Subgrade Stabilization):** To address the prevalent issue of weak subgrade soils, it is highly recommended to prioritise chemical stabilisation using cost-effective industrial by-products (e.g., lime or fly ash). This approach substantially increases the California Bearing Ratio (CBR), providing a rigid platform that reduces the required thickness of overlying flexible layers.

2. **Surface Level (Optimized Bituminous Mix):** The Roads and Highways Department (RHD) must revise existing material specifications to enforce rigorous quality control on imported bitumen and mandate aggregate shape sorting. Specifically, introducing a specification for a minimum threshold of 20% cubical coarse aggregates paired with local ERL bitumen will fundamentally enhance the interlocking matrix.

Implementing these two policy shifts will substantially increase the structural layer coefficient (a_1) of the surfacing course. This integrated approach reduces the total volume of imported materials required, thereby delivering significant lifecycle cost savings and structural resilience for national highway projects.

5.3 Future Work

Due to laboratory constraints and project limitations, several crucial performance tests could not be executed within the scope of this study. The following areas are proposed for future investigation:

- **Indirect Tensile Strength (ITS) Testing:** Conducting ITS tests (ASTM D6931) to evaluate the tensile strength characteristics and fracture resistance of the mix combinations under loading.
- **Moisture Susceptibility and Climate Resistance:** Performing the Retained Stability test (Marshall immersion) or Tensile Strength Ratio (TSR) tests to evaluate the long-term durability and stripping resistance of the mixes under conditions representing Bangladesh's subtropical monsoon climate.
- **Rutting and Fatigue Evaluation:** Conducting the Hamburg Wheel Tracking Test (AASHTO T 324) to evaluate empirical rutting depth progression, and the Four-Point Bending Beam Fatigue Test (AASHTO T 321) to assess the fatigue cracking resistance of the bituminous mixtures under cyclic traffic loading.

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